DO NOW — RECALL

Lesson 126 · Addressing Modes: Immediate, Direct, Indi

Name: _____ Class: _____ Date: _____

1. Which LMC instruction reads a mailbox value into the accumulator?

2. In a BST, which side holds a smaller value?

3. How many bits are in three bytes?

4. What is a function parameter?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

LDA (load).

2. Model answer

The left subtree.

3. Model answer

24.

4. Model answer

An input used by a subroutine.

By the end, you can…

• LOAD #12 uses literal 12; LOAD 12 uses contents of memory[12].

• LOAD @12 uses address stored in memory[12]; then loads data there.

• LOAD 20,X uses effective address 20+X; useful for array elements.

Vocabulary — know these terms

immediate — Operand is a literal value.

direct — Operand gives the memory location holding a value.

indirect — Operand points to a location holding the effective address.

indexed — Base address plus index gives effective address.

NEW LEARNING

Read, annotate and explain

Immediate versus direct

LOAD #12 uses literal 12; LOAD 12 uses contents of memory[12].

Indirect addressing

LOAD @12 uses address stored in memory[12]; then loads data there.

Indexed addressing

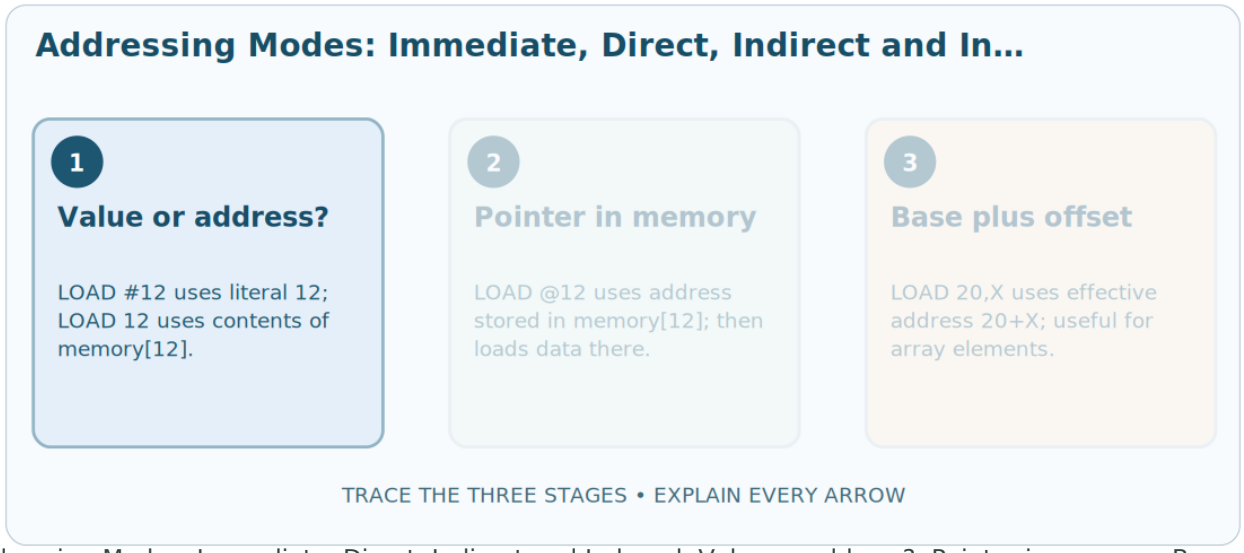
LOAD 20,X uses effective address 20+X; useful for array elements.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

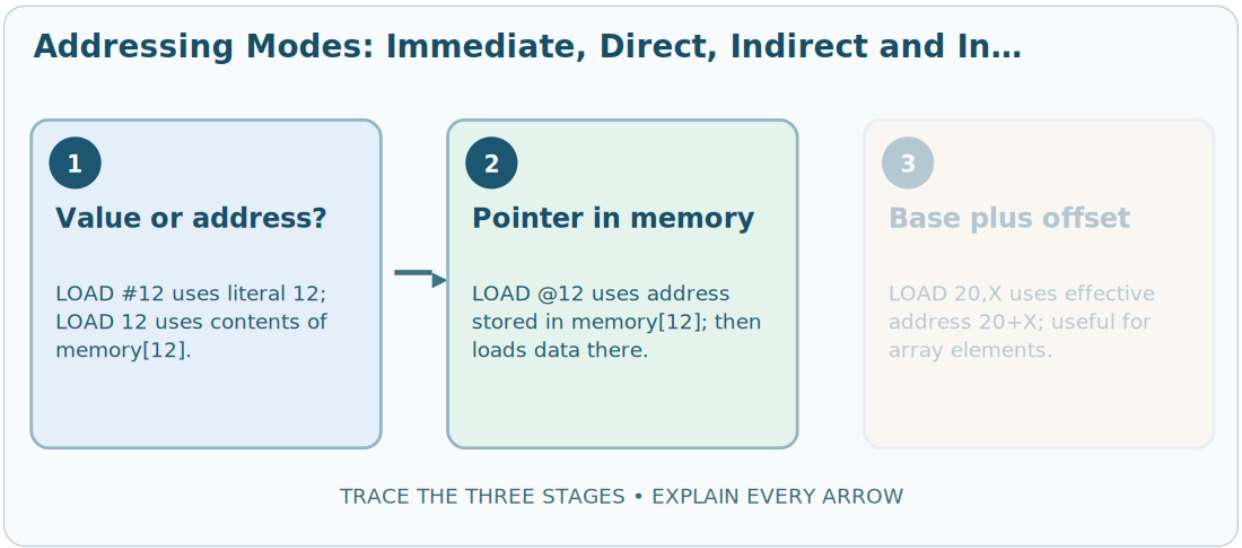
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Addressing Modes: Immediate, Direct, Indirect and Indexed: Value or address?; Pointer in memory; Base plus offset

New Learning

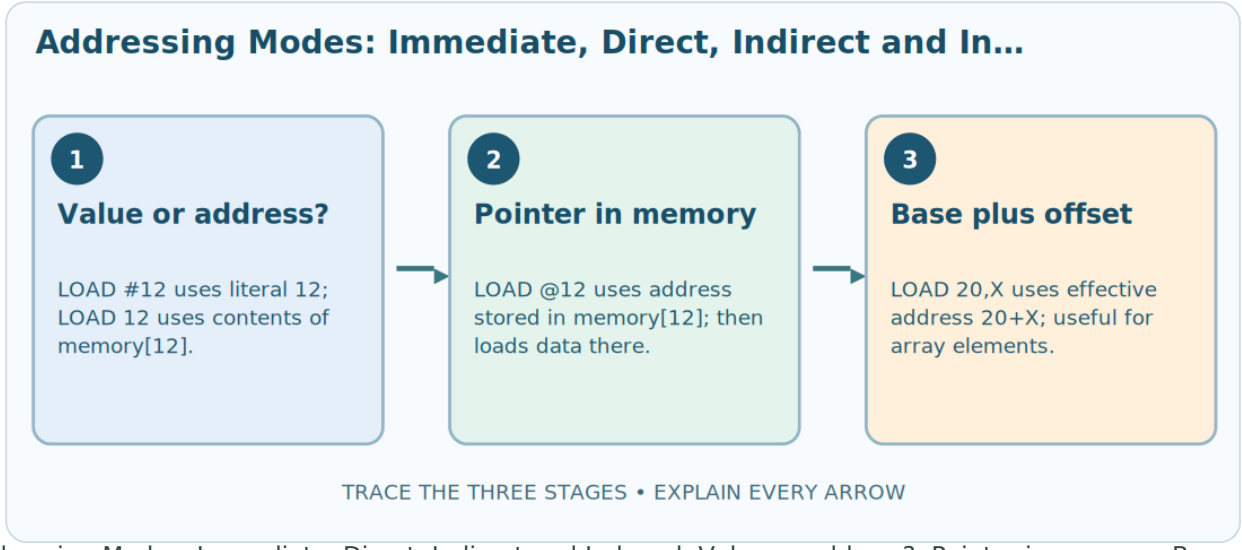


Addressing Modes: Immediate, Direct, Indirect and Indexed: Value or address?; Pointer in memory; Base plus offset

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Addressing Modes: Immediate, Direct, Indirect and Indexed: Value or address?; Pointer in memory; Base plus offset

I DO — WORKED EXAMPLE

Follow the worked method

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

#12 gives literal 12

Predict the first step from the prompt.

12 reads memory[12]=40

Explain the intermediate result or decision.

@12 follows pointer to 7

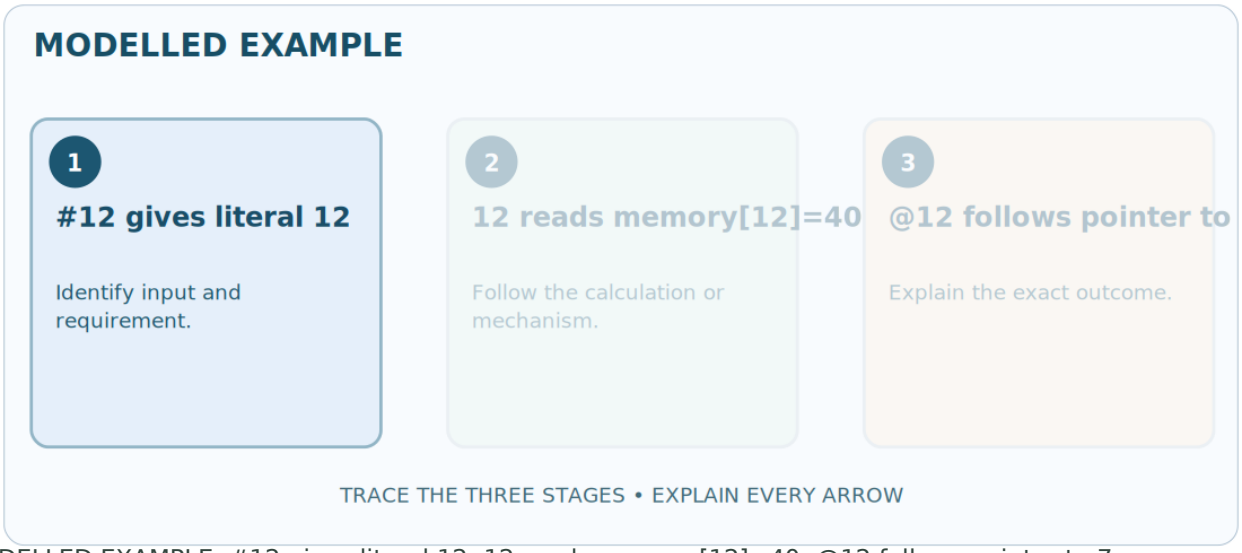
Using illustrative assembler conventions: immediate gives 12; direct reads 40; indirect follows memory[12]=40 then loads memory[40]=7.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

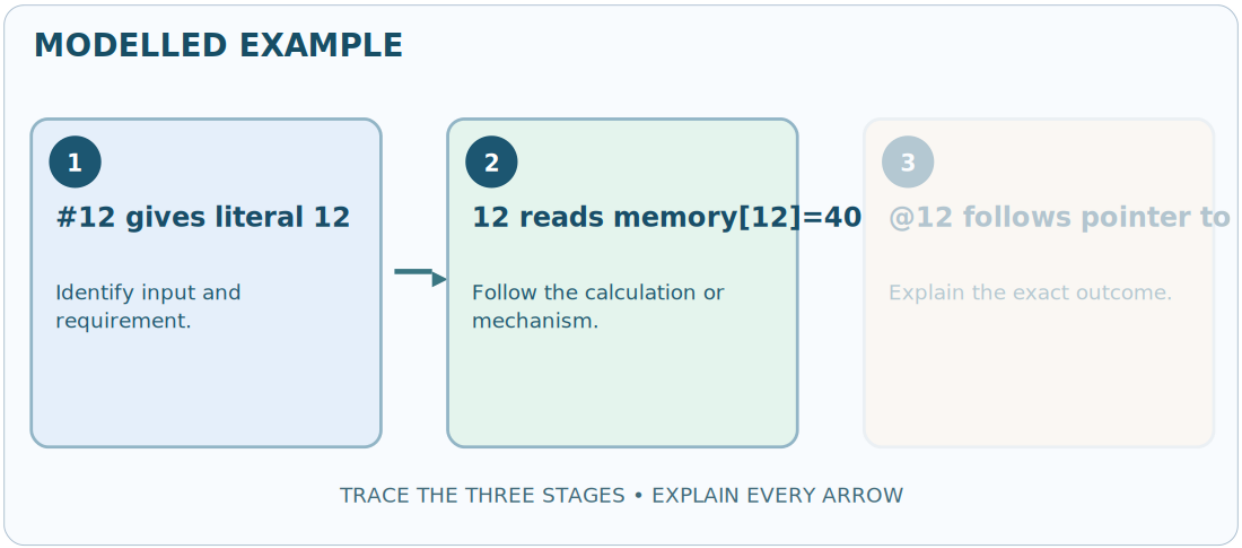
Predict the first step from the prompt.



MODELLED EXAMPLE: #12 gives literal 12; 12 reads memory[12]=40; @12 follows pointer to 7

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

Predict the first step from the prompt.



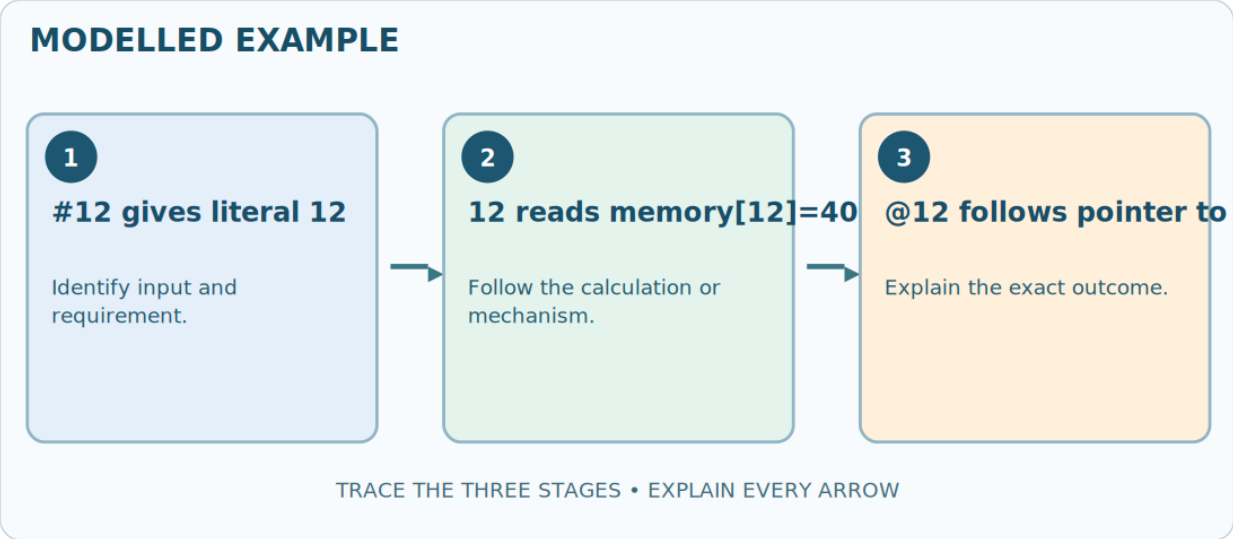
MODELLED EXAMPLE: #12 gives literal 12; 12 reads memory[12]=40; @12 follows pointer to 7

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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MODELLED EXAMPLE: #12 gives literal 12; 12 reads memory[12]=40; @12 follows pointer to 7

WE DO — GUIDED PRACTICE

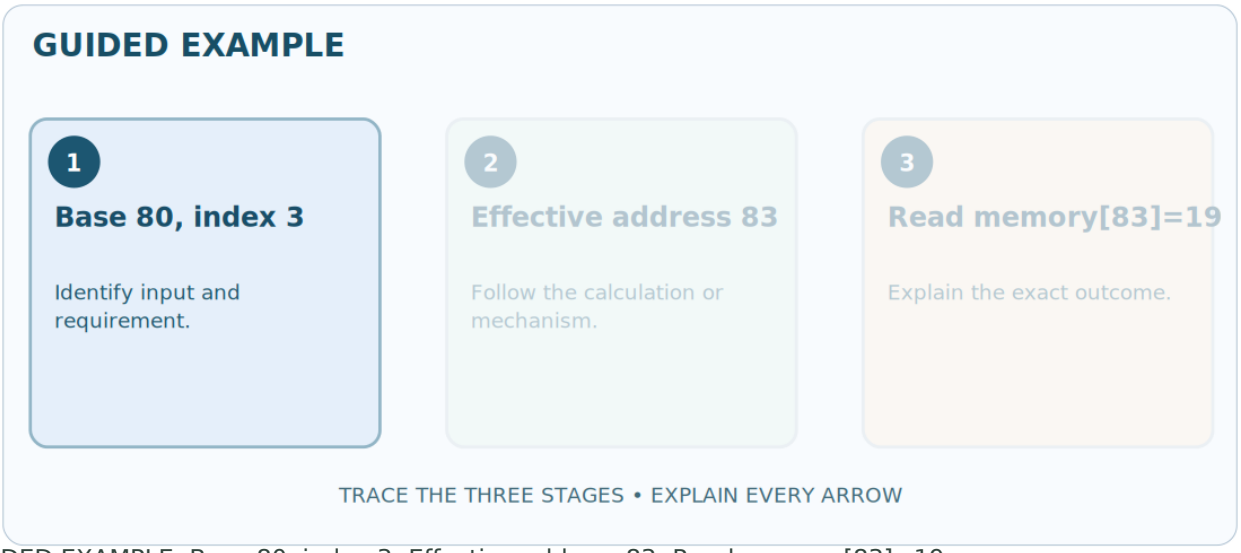
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Array starts at memory address 80. X=3 and memory[83]=19. What does LOAD 80,X load?

Predict the first step from the prompt.



GUIDED EXAMPLE: Base 80, index 3; Effective address 83; Read memory[83]=19

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Base 80, index 3 Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define immediate and direct addressing with one example each. [6]

Question 2

Memory[15]=29 and memory[29]=6. Give values loaded by direct LOAD 15 and indirect LOAD @15. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain indexed addressing for an array at base 50 with index 4 and discuss why this is not simply LMC LDA notation. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In direct addressing, the operand always means the literal value to be used.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

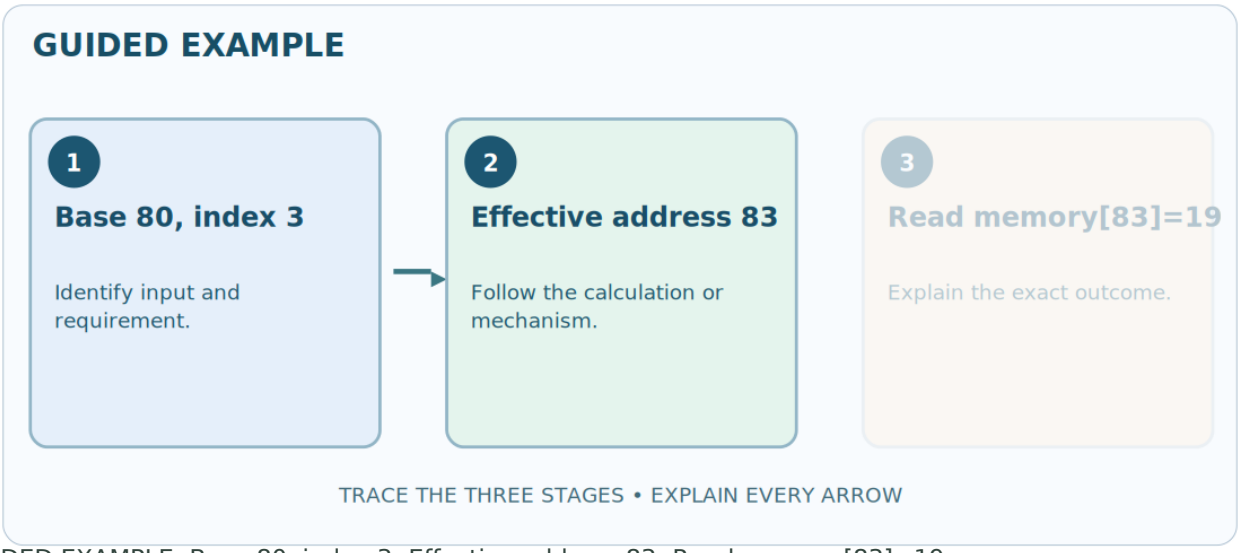
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Array starts at memory address 80. X=3 and memory[83]=19. What does LOAD 80,X load?

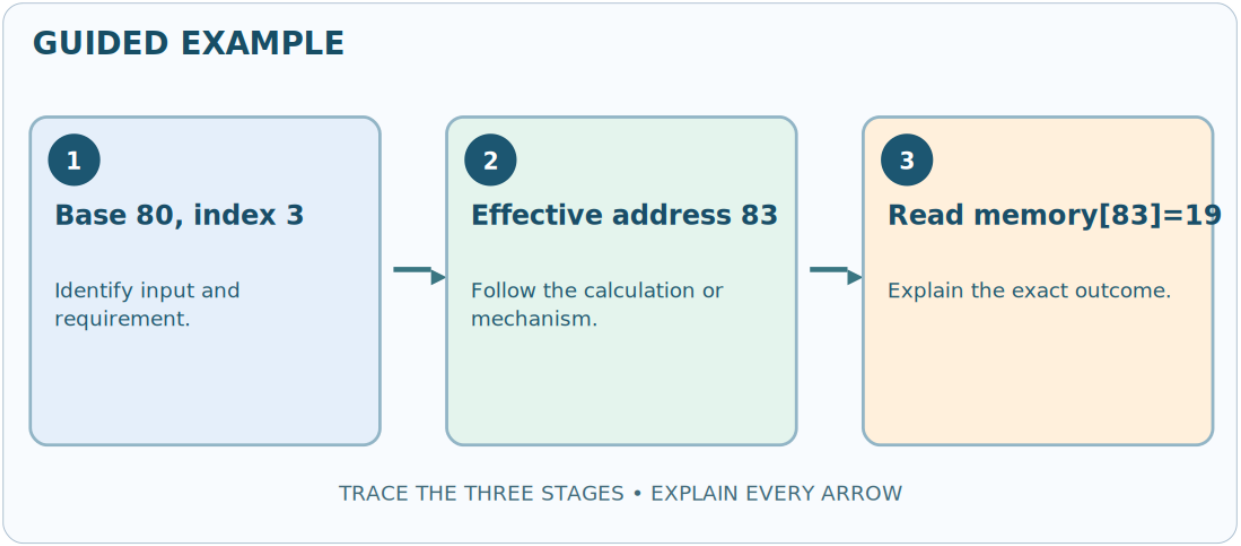
Predict the first step from the prompt.



GUIDED EXAMPLE: Base 80, index 3; Effective address 83; Read memory[83]=19

Array starts at memory address 80. X=3 and memory[83]=19. What does LOAD 80,X load?

Predict the first step from the prompt.



GUIDED EXAMPLE: Base 80, index 3; Effective address 83; Read memory[83]=19

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Effective address 83

Explain the intermediate result or decision.

Read memory[83]=19

Effective address = 80+3=83. It loads 19. Indexed addressing avoids writing a new absolute address for each element.

IMPROVE: Explain one correction to your own reasoning.